Exam 1 MTH 207 (Fall 2026) — J. Baggett

Score / 100

First Name	Last Name

Before you start. Write your name above and check that you have all 6 pages. All electronics off and out of sight — phones included, even just to tell the time, even in airplane mode. No calculators, notes, or books.

TO EARN POINTS YOU MUST SHOW YOUR WORK. Explain your reasons, justify your steps, and present your logic in an organized way. An answer with no supporting work earns no credit.

You have exactly 55 minutes. When you are told to stop you must stop **immediately** — that is in fairness to everyone. Keep your eyes on your own paper.

(2^{pts} ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) If the average rate of change of f on $[1, 3]$ is 0, then f is constant on $[1, 3]$.

(b) $\lim_{x\to 3} \dfrac{x^2 - 9}{x - 3} = \lim_{x\to 3}(x + 3)$

(c) If $\lim_{x\to 0} f(x) = \infty$ and $\lim_{x\to 0} g(x) = \infty$, then $\lim_{x\to 0}[f(x) - g(x)] = 0$.

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(10^{pts}) **2.** Evaluate the following limits. As a reminder, **justify your steps!** Simplify your answer as much as possible. (If the limit does not exist, state why.)

(a) $\lim_{x\to 3} \dfrac{\frac{1}{x}-\frac{1}{3}}{x-3}$

(b) Suppose $2x \le f(x) \le x^2 - 2x + 4$ for all x. Find $\lim_{x\to 2} f(x)$.

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(13^{pts}) **3.** Evaluate the following limits. As a reminder, **justify your steps!** (If the limit does not exist, state why.)

(a) $\lim_{x \to 2} \dfrac{x^2 - 4}{|x - 2|}$

(b) $\lim_{x \to 2^+} \dfrac{x - 5}{x^2 - 4}$, and then say what your answer tells you about the graph of this function.

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(7^{pts}) **4.** State the Intermediate Value Theorem.

/7

(11^{pts}) **5.** Find the value of the constant c that makes the function below continuous at every real number. Justify your answer.

$$f(x) = \begin{cases} \dfrac{x^2 + x - 6}{x - 2} & x \neq 2 \\ c & x = 2 \end{cases}$$

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(9^{pts}) **6.** Find the horizontal asymptotes of $f(x) = \dfrac{6 - 4x^2}{3 + 2x^2}$.

/9

(11^{pts}) **7.** Evaluate the following limit or state that it does not exist. Show the steps!

$$\lim_{x\to-\infty}\left(\sqrt{4x^2+3x}+2x\right)$$

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(15^{pts}) **8.** Let $f(x)=\sqrt{x+1}$. Use the **definition of derivative** to compute $f'(3)$, and then say what that number represents. Using shortcut rules will earn no points.

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(7^{pts}) **9.** Give an example of a function f for which $\lim_{x\to 0} f(x)$ does **not** exist but $\lim_{x\to 0} |f(x)|$ **does** exist. Explain why your example works.

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(11^{pts}) **10.** Let $g(x) = x^3 - 4x^2 + 3x + 2e^x$. Find $g'(x)$, then find the equation of the tangent line to $y = g(x)$ at $x = 0$.

/11

THIS IS THE LAST PAGE OF THE EXAM.